EXP 18: Implement Perceptron based IRIS classification

CODING:

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# Perceptron-Based Iris Classification

# Training data: [Sepal Length, Sepal Width]
X = [
    [5.1, 3.5],
    [4.9, 3.0],
    [5.0, 3.4],
    [6.4, 3.2],
    [6.9, 3.1],
    [5.5, 2.3]
]

# Setosa = 1, Non-Setosa = 0
Y = [1, 1, 1, 0, 0, 0]

# Initialize weights and bias
w1 = 0
w2 = 0
bias = 0
learning_rate = 0.1

# Training the perceptron
for epoch in range(20):
    for i in range(len(X)):
        z = w1 * X[i][0] + w2 * X[i][1] + bias
```

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prediction = 1 if z >= 0 else 0


error = Y[i] - prediction


w1 += learning_rate * error * X[i][0]

w2 += learning_rate * error * X[i][1]

bias += learning_rate * error

# Test a new flower

test = [5.0, 3.4]


z = w1 * test[0] + w2 * test[1] + bias


result = 1 if z >= 0 else 0


print("IRIS FLOWER CLASSIFICATION")

print("-------------------------")

print("Sepal Length:", test[0])

print("Sepal Width:", test[1])


if result == 1:

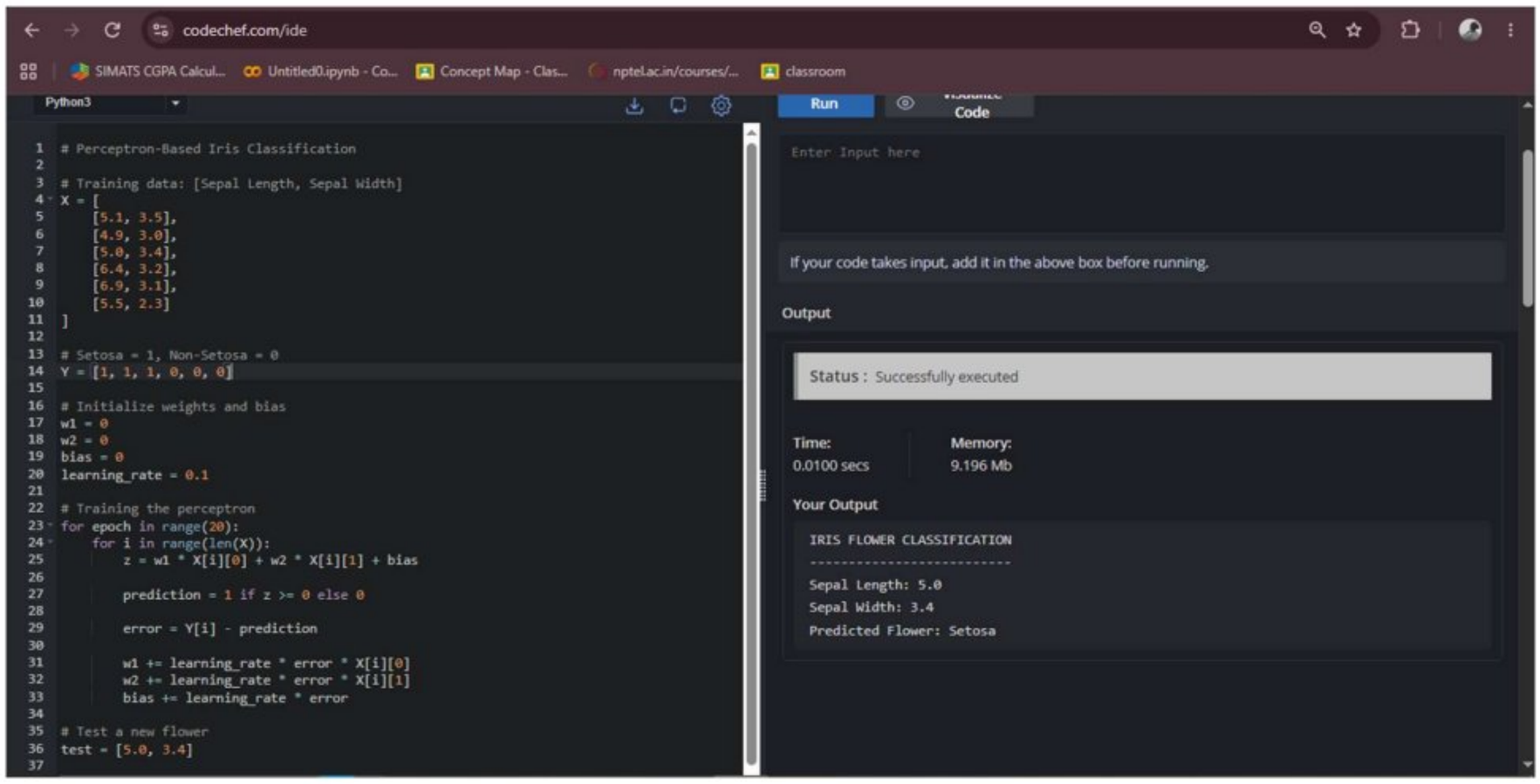
   print("Predicted Flower: Setosa")

else:

   print("Predicted Flower: Non-Setosa")
```

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OUTPUT:



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